

Matching Estimators and Linear Regression

Angrist (1998): Proof for the Saturated and General Linear Cases

ECON 712 — Supplementary Notes

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1 Setup and Notation

Let $D_i \in \{0, 1\}$ be a binary treatment, X_i a vector of pre-treatment covariates, and $\{Y_{0i}, Y_{1i}\}$ the potential outcomes. The observed outcome is $Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$.

Define the following population objects:

$$\begin{aligned} p(x) &= E[D_i \mid X_i = x] && \text{(propensity score)} \\ \mu_0(x) &= E[Y_{0i} \mid X_i = x] && \text{(baseline outcome CEF)} \\ \delta(x) &= E[Y_{1i} - Y_{0i} \mid X_i = x] && \text{(conditional ATE)} \end{aligned}$$

The true data-generating process can therefore be written as

$$Y_i = \mu_0(X_i) + \delta(X_i) D_i + u_i, \quad E[u_i \mid D_i, X_i] = 0, \quad (1)$$

where the zero-mean condition on u_i follows from the **Conditional Independence Assumption (CIA)**: $(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i \mid X_i$.

What we want to know. We run a regression that *linearly controls* for X_i — either a fully saturated specification or a plain linear-in- X specification. In both cases we use the Frisch–Waugh–Lovell (FWL) theorem to isolate the coefficient on D_i . The question is: what causal parameter does that coefficient identify under CIA?

2 The Frisch–Waugh–Lovell Starting Point

Consider any regression of the form

$$Y_i = \alpha + \delta D_i + g(X_i) + \varepsilon_i$$

for some control function $g(\cdot)$. By FWL, the population coefficient on D_i is

$$\text{plim } \hat{\delta} = \frac{E[\tilde{D}_i Y_i]}{E[\tilde{D}_i^2]}, \quad (2)$$

where $\tilde{D}_i = D_i - \hat{p}(X_i)$ and $\hat{p}(X_i) = L[D_i \mid g(X_i)]$ is the linear projection of D_i on the control function.

The two cases we study differ *only* in what $\hat{p}$ is:

- **Saturated regression:** $g(X_i)$ includes a dummy for every cell of X_i , so the projection is the conditional expectation: $\hat{p}^{\text{sat}}(X_i) = p(X_i)$ exactly.
- **Linear regression:** $g(X_i) = \beta' X_i$, so $\hat{p}^{\text{lin}}(X_i) = \Pi_0 + \Pi' X_i$ is only the best linear approximation to $p(X_i)$; in general $\hat{p}^{\text{lin}} \neq p$.

3 The Saturated Case

This is the proof given (implicitly) by Angrist (1998) and made explicit in Angrist and Pischke (2009, Ch. 3).

When the regression is saturated, $\tilde{D}_i^{\text{sat}} = D_i - p(X_i)$.

Denominator

$$E[(\tilde{D}_i^{\text{sat}})^2] = E[(D_i - p(X_i))^2] \stackrel{\text{LIE}}{=} E[\text{Var}(D_i \mid X_i)].$$

Numerator: Term A — baseline outcomes

$$E[\tilde{D}_i^{\text{sat}} Y_{0i}] = E[E[(D_i - p(X_i)) Y_{0i} \mid X_i]].$$

Under CIA, $Y_{0i} \perp D_i \mid X_i$, so

$$E[(D_i - p(X_i)) Y_{0i} \mid X_i] = \underbrace{E[D_i - p(X_i) \mid X_i]}_{=0} \cdot E[Y_{0i} \mid X_i] = 0.$$

Numerator: Term B — treatment effect heterogeneity

Under CIA $E[\delta_i \mid D_i, X_i] = \delta(X_i)$, so the $(\delta_i - \delta(X_i))D_i$ residual satisfies $E[\tilde{D}_i \cdot (\delta_i - \delta(X_i))D_i \mid D_i, X_i] = 0$. Thus we only need:

$$E[\tilde{D}_i^{\text{sat}} \delta(X_i) D_i] = E[\delta(X_i) E[(D_i - p(X_i)) D_i \mid X_i]].$$

For binary D_i , $D_i^2 = D_i$, so

$$E[(D_i - p(X_i)) D_i \mid X_i] = p(X_i) - p(X_i)^2 = p(X_i)(1 - p(X_i)) = \text{Var}(D_i \mid X_i).$$

Hence

$$E[\tilde{D}_i^{\text{sat}} \delta(X_i) D_i] = E[\delta(X_i) \text{Var}(D_i \mid X_i)].$$

Result

Assembling (2):

Proposition 1 (Angrist 1998 — saturated regression). *Under CIA and a fully saturated regression,*

$$\text{plim } \hat{\delta}^{\text{sat}} = \frac{E[\delta(X_i) \text{Var}(D_i \mid X_i)]}{E[\text{Var}(D_i \mid X_i)]} = \int \delta(x) \omega(x) dx,$$

where the variance weights are

$$\omega(x) = \frac{\text{Var}(D_i \mid X_i = x) \cdot f(x)}{E[\text{Var}(D_i \mid X_i)]}, \quad \int \omega(x) dx = 1.$$

Remark 1. The weights $\omega(x)$ are largest where the propensity score is closest to $\frac{1}{2}$, i.e. where treatment varies most. Under CIA, $\text{ATT}(x) = \text{ATU}(x) = \text{ATE}(x) = \delta(x)$, so $\hat{\delta}^{\text{sat}}$ is a weighted average of the ATE across covariate cells.

4 The General Linear Case

Now suppose we only control linearly: $Y_i = \alpha + \delta D_i + \beta' X_i + \varepsilon_i$. The projected propensity score is $\hat{p}^L(X_i) = \Pi_0 + \Pi' X_i$, the best linear predictor of D_i given $(1, X_i)$.

Key object. Define the *propensity-score misspecification residual*

$$\lambda(X_i) \equiv p(X_i) - \hat{p}^L(X_i). \tag{3}$$

This is the component of the propensity score that the linear regression *cannot capture* — the nonlinearity in $p(\cdot)$. The OLS normal equations imply $E[\lambda(X_i)] = 0$ and $E[\lambda(X_i) X_i] = \mathbf{0}$.

The linear regression residual decomposes as

$$\tilde{D}_i^{\text{lin}} = D_i - \hat{p}^L(X_i) = \underbrace{(D_i - p(X_i))}_{e_i} + \lambda(X_i), \quad E[e_i \mid X_i] = 0.$$

4.1 Denominator

$$\begin{aligned}
E[(\tilde{D}_i^{\text{lin}})^2] &= E[e_i^2] + 2 E[e_i \lambda(X_i)] + E[\lambda(X_i)^2] \\
&= E[\text{Var}(D_i | X_i)] + 2 E \left[\underbrace{E[e_i | X_i]}_{=0} \lambda(X_i) \right] + E[\lambda(X_i)^2] \\
&= E[\text{Var}(D_i | X_i)] + \text{Var}(\lambda(X_i)).
\end{aligned}$$

The denominator is *strictly larger* than in the saturated case whenever $\lambda \neq 0$.

4.2 Numerator

From (1), $Y_i = \mu_0(X_i) + \delta(X_i)D_i + u_i$.

Term A: baseline outcomes.

$$\begin{aligned}
E[\tilde{D}_i^{\text{lin}} Y_{0i}] &= E \left[E[(D_i - \hat{p}^L(X_i)) Y_{0i} | X_i] \right] \\
&= E \left[(E[D_i | X_i] - \hat{p}^L(X_i)) \mu_0(X_i) \right] \quad (\text{CIA: } Y_{0i} \perp D_i | X_i) \\
&= E[\lambda(X_i) \mu_0(X_i)].
\end{aligned}$$

Unlike the saturated case, this is *not* zero unless $\lambda \equiv 0$.

Term B: treatment effects.

The residual $(\delta_i - \delta(X_i))D_i$ contributes zero by the same CIA argument as before. For the systematic part:

$$\begin{aligned}
E[\tilde{D}_i^{\text{lin}} \delta(X_i) D_i] &= E \left[\delta(X_i) E[(D_i - \hat{p}^L(X_i)) D_i | X_i] \right] \\
&= E \left[\delta(X_i) p(X_i) (1 - \hat{p}^L(X_i)) \right].
\end{aligned}$$

Decompose $1 - \hat{p}^L = (1 - p) + \lambda$:

$$= E[\delta(X_i) \text{Var}(D_i | X_i)] + E[\delta(X_i) p(X_i) \lambda(X_i)].$$

Combining. Adding Terms A and B:

$$E[\tilde{D}_i^{\text{lin}} Y_i] = E[\delta(X_i) \text{Var}(D_i | X_i)] + E \left[\lambda(X_i) \underbrace{(\mu_0(X_i) + \delta(X_i)p(X_i))}_{= E[Y_i | X_i]} \right].$$

Simplifying the bias term. Decompose $E[Y_i | X_i] = L[E[Y_i | X_i] | 1, X_i] + \rho(X_i)$, where $\rho(X_i)$ is the *nonlinear residual of the outcome CEF*. Since $\lambda(X_i)$ is orthogonal to all linear functions of $(1, X_i)$ (by the normal equations),

$$E[\lambda(X_i) E[Y_i | X_i]] = \underbrace{E[\lambda(X_i) L[E[Y_i | X_i] | 1, X_i]]}_{=0} + E[\lambda(X_i) \rho(X_i)] = \text{Cov}(\lambda(X_i), \rho(X_i)).$$

4.3 Result

Proposition 2 (Linear regression — general case). *Under CIA and the linear regression $Y_i = \alpha + \delta D_i + \beta' X_i + \varepsilon_i$,*

$$\text{plim } \hat{\delta}^{\text{lin}} = \frac{E[\delta(X_i) \text{Var}(D_i | X_i)] + \text{Cov}(\lambda(X_i), \rho(X_i))}{E[\text{Var}(D_i | X_i)] + \text{Var}(\lambda(X_i))}, \quad (4)$$

where

$$\begin{aligned} \lambda(X_i) &= p(X_i) - L[p(X_i) | 1, X_i] && (\text{nonlinear part of propensity score}), \\ \rho(X_i) &= E[Y_i | X_i] - L[E[Y_i | X_i] | 1, X_i] && (\text{nonlinear part of outcome CEF}). \end{aligned}$$

5 Comparing the Two Cases

5.1 Structural comparison

	Saturated	Linear
$\hat{p}(X_i)$	$p(X_i)$	$L[p(X_i) 1, X_i]$
$\lambda(X_i)$	0	$p(X_i) - \hat{p}^L(X_i)$
Denominator	$E[\text{Var}(D_i X_i)]$	$E[\text{Var}(D_i X_i)] + \text{Var}(\lambda)$
Numerator bias	0	$\text{Cov}(\lambda(X_i), \rho(X_i))$
Weights sum to 1?	Yes	Not in general

5.2 When does the linear case recover the variance-weighted formula?

Setting $\lambda \equiv 0$ in (4) immediately yields Proposition 1. Three sufficient conditions for $\lambda \equiv 0$ (or for the bias terms to vanish) are:

1. **The propensity score is linear in X_i .** Then $\hat{p}^L(X_i) = p(X_i)$ exactly and $\lambda \equiv 0$. The linear regression gives the same result as the saturated regression *regardless* of whether $\mu_0(X_i)$ is linear.
2. **The outcome CEF is linear in X_i .** Then $\rho \equiv 0$, so $\text{Cov}(\lambda, \rho) = 0$. *But* the denominator is still inflated by $\text{Var}(\lambda) \geq 0$ unless $\lambda \equiv 0$ as well. The pure linear-outcome case does *not* by itself recover the variance-weighted formula.
3. **Constant treatment effects and linear outcome.** If $\delta(x) = \delta$ for all x and $E[Y_i | X_i]$ is linear, a calculation shows that the numerator bias equals $\delta \text{Var}(\lambda)$, so the denominator inflation and numerator bias exactly cancel:

$$\text{plim } \hat{\delta}^{\text{lin}} = \frac{\delta E[\text{Var}(D_i | X_i)] + \delta \text{Var}(\lambda)}{E[\text{Var}(D_i | X_i)] + \text{Var}(\lambda)} = \delta.$$

This is why a (correctly specified) linear regression with linear controls is consistent for the constant ATE even when the propensity score is nonlinear.

5.3 Interpreting the bias

The bias term $\text{Cov}(\lambda(X_i), \rho(X_i))$ has an intuitive reading:

- $\lambda(X_i)$ is the part of the propensity score the linear regression *fails to absorb* — cells where the true treatment probability curves away from the fitted linear index.
- $\rho(X_i)$ is the part of the outcome function the linear regression *fails to absorb* — cells where $E[Y_i | X_i]$ is curved.
- The bias arises when these two forms of misspecification are *correlated*: the cells where the linear regression gets the propensity score wrong tend to be the same cells where it also gets the outcome function wrong. OLS then conflates the two, distorting the treatment-effect estimate.

5.4 Why the saturated case avoids the problem

In the saturated case, $g(X_i)$ contains a dummy for every cell of X_i , so the linear projection of D_i on $g(X_i)$ is the cell mean $p(X_i)$. The regression is therefore also saturating the propensity-score model. By construction $\lambda \equiv 0$, which kills both the numerator bias and the denominator inflation. There is no residual nonlinearity for ρ to correlate with.

6 Summary

Theorem 3 (Regression–Matching equivalence). *Under CIA:*

- Saturated regression** always identifies the variance-weighted ATE of Proposition 1, with weights proportional to the conditional variance of treatment.
- Linear regression** identifies the same quantity if and only if $\lambda(X_i) \equiv 0$, i.e. the true propensity score $p(X_i)$ is linear in X_i . Otherwise, the estimand differs by the correction terms in Proposition 2.
- In either case, if treatment effects are constant ($\delta(x) = \delta$) and the outcome equation is correctly specified, the regression identifies the constant ATE = ATT = ATU = δ .

The practical implication is that with heterogeneous treatment effects and an unknown propensity score, a saturated regression (or a matching estimator using the nonparametric $p(X_i)$) gives a cleaner and more interpretable weighted average than a plain linear regression, which may produce weights that are difficult to characterize and that can even be negative when $p(X_i)$ is strongly nonlinear.

Reference. Angrist, J.D. (1998). “Estimating the Labor Market Impact of Voluntary Military Service Using Social Security Data on Military Applicants.” *Econometrica*, 66(2), 249–288.

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